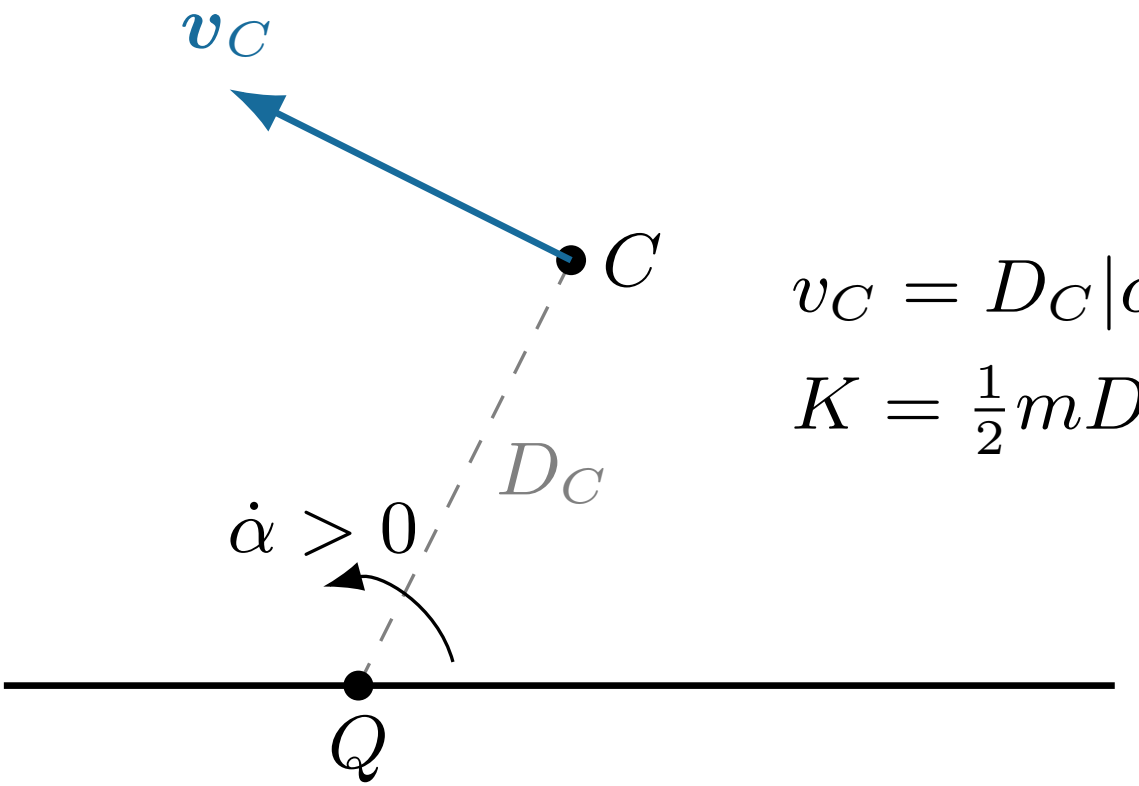

$$v_C = D_C|\dot{\alpha}|$$

$$K = \tfrac{1}{2} m D_C^2 \dot{\alpha}^2$$